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Assessment of the efficacy of an integrated rehabilitation program in relapse prevention among opioid use disorder patients in psychiatry and neurology center, Tanta university

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Abstract

Background Relapse remains a major challenge in substance use disorder (SUD). Cognitive-behavioral therapy (CBT) and mindfulness-based cognitive therapy (MBCT) are effective in reducing relapse and improving quality of life (QoL) for individuals with SUD. This study aimed to develop, culturally adapt, and evaluate the feasibility and preliminary efficacy of an integrated psychotherapeutic program for individuals with opioid use disorder (OUD) in Egypt, and to identify predictors of relapse.

Methods A quasi-experimental study was conducted to compare the effectiveness of either an integrated intervention combining CBT-based relapse prevention, MI, MBCT, family psychoeducation, peer support, and twelve-step facilitation, or treatment-as-usual (TAU). Feasibility outcomes included recruitment, retention, and adherence. Clinical outcomes relapse rates, Addiction Severity Index (ASI) scores, and WHOQOL-BREF scores were assessed at baseline, 3 months, and 6 months. Predictors of relapse were examined using Cox regression.

Results The integrated program showed high feasibility, with 90% retention and strong adherence, and demonstrated clear superiority over TAU. Relapse rates were significantly lower at 3 and 6 months (33.3% vs. 66.7% and 44.4% vs. 77.8%; $p=0.001$). Participants exhibited significant reductions in all ASI domains and all WHOQOL-BREF dimensions. Survival analysis confirmed a substantially lower relapse hazard, while multivariate modelling identified unemployment, illiteracy, unmarried status, higher opioid dose, repeated detoxification attempts, and poor adherence as independent predictors of relapse.

Conclusions The culturally adapted, integrated psychotherapeutic program is a feasible and promising intervention, associated with reduced relapse rates, significant improvements in addiction severity, and enhanced quality of life over six months. These findings support the implementation of culturally tailored, multimodal psychological interventions for OUD in low- and middle-income countries like Egypt.

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Introduction

Substance use disorders (SUDs) are characterized by maladaptive patterns of substance use that lead to significant clinical impairment, recurrent relapse, and substantial social and economic burdens [1]. Among SUDs, opioid use disorder (OUD) represents a critical public health challenge, encompassing both illicit opioids, such as heroin, and prescription analgesics, including oxycodone, codeine, and morphine [2].

In Egypt, epidemiological data reveal a concerning prevalence of substance use. Surveys among secondary school students indicate lifetime use rates of 5.05% for cannabis and 0.84% for opium [3]. Furthermore, the 2023 National Survey of Psychiatric and Substance Use Disorders estimated that 5.4% of the population met the criteria for a SUD, with opioids accounting for 39.5% of these cases [3]. These findings underscore the urgent need for evidence-based, culturally adapted interventions for OUD in this context.

Globally, psychological interventions such as cognitive-behavioral therapy (CBT), motivational interviewing (MI), and mindfulness-based cognitive therapy (MBCT) have demonstrated efficacy in reducing substance use, preventing relapse, and improving psychosocial functioning [4–7]. CBT and MI target motivation, problem-solving skills, and emotional regulation, thereby mitigating substance use and enhancing social and occupational outcomes [5, 6]. MBCT integrates mindfulness practices with cognitive-behavioral strategies to modify maladaptive thought patterns and bolster relapse prevention [7, 8].

Despite robust international evidence for these modalities, their integration into a culturally adapted, multimodal framework remains largely unexplored in Egypt. Current rehabilitation services predominantly rely on pharmacotherapy or non-standardized psychosocial approaches [9]. Additionally, quality of life (QoL), a key indicator of recovery, is often underutilized as an outcome measure in SUD research within these settings [10].

This study therefore aimed to develop and evaluate a culturally tailored, integrated rehabilitation program for patients with OUD in Egypt, combining CBT, MI, MBCT, family psychoeducation, peer support, and twelve-step facilitation. A further objective was to identify clinical, psychological, and sociodemographic predictors of relapse and to assess changes in QoL. The study seeks to provide context-specific evidence to guide comprehensive addiction treatment in resource-limited settings.

Patients and methods

Study design and setting

This quasi-experimental, controlled, parallel-group study with a 6-month follow-up was conducted at the Psychiatry, Neurology, and Neurosurgery Center of Tanta University, Egypt, between June 2024 and May 2025. Participants were allocated to receive either a culturally adapted, integrated psychotherapeutic program or treatment-as-usual (TAU). Outcome measures were assessed at baseline, 3 months (post-intervention), and 6 months.

Participants

Male inpatients aged 18–45 years diagnosed with OUD according to DSM-5 criteria were eligible if they were medically stable after detoxification, able to participate in structured sessions, and provided informed consent. Exclusion criteria included active psychotic, manic, or panic disorders; severe medical illness (e.g., infective endocarditis, HIV, hepatitis); cognitive impairment interfering with participation; or inability to commit to follow-up assessments.

Sample size

Sample size was calculated using Epi Info 7, assuming a 95% confidence level, 80% power, and expected relapse rates of 23% in the intervention group versus 45% in controls [3]. Accounting for 10% attrition, 50 participants per group (total $N = 100$) were enrolled.

Recruitment and allocation

Eligible patients admitted for detoxification were screened consecutively. Participants were allocated to groups based on treatment-unit availability to maintain naturalistic clinical conditions, consistent with a quasi-experimental design. All participants provided written informed consent prior to enrollment.

Interventions

A- integrated psychotherapeutic program

The intervention was a structured, culturally adapted, multimodal Psychotherapeutic rehabilitation program integrating.

1. Motivational interviewing (MI).
2. CBT-based relapse prevention.
3. Mindfulness-based cognitive therapy (MBCT).
4. Family psychoeducation.
5. Peer support groups.
6. Twelve-step facilitation.

The program comprised 24 group sessions and 4 family sessions delivered over approximately 6 months. Sessions lasted 90 min, conducted twice weekly by trained therapists using standardized manuals. Structured homework reinforced session content. Groups were closed (8–12 participants) and co-facilitated by two therapists.

B- Treatment-as-Usual (Control Group)

TAU included medically supervised detoxification, pharmacotherapy with oral naltrexone, and supportive counseling focused on abstinence maintenance and stress management. TAU did not include structured psychotherapeutic modules, mindfulness training, or family/peer interventions.

Outcome measures

A- primary outcome

Relapse, defined as opioid-positive urine toxicology at 3 or 6 months, cross-validated with self-report.

B- Secondary Outcomes

1. Addiction Severity Index (ASI, 5th edition; Arabic version) assessing medical, legal, occupational, substance use, social, and psychiatric domains at baseline and 6 months [9].
2. Quality of Life (WHOQOL-BREF, Arabic version) assessing physical, psychological, social, and environmental domains at baseline, 3 months, and 6 months. Internal consistency in a pilot sample: Cronbach's $\alpha = 0.781$ [10].

C- additional measures

Sociodemographic characteristics (age, gender, marital status, education, employment), daily opioid dose (< 1 g vs. ≥ 1 g), duration of addiction, previous detoxification attempts, program attendance ($> 60\%$ vs. $\leq 60\%$), and homework adherence ($\geq 70\%$ vs. $< 70\%$) were recorded as additional measures to explore potential predictors of relapse and treatment outcomes [11].

Procedures

All participants underwent a standardized detoxification protocol before baseline assessment. Semi-structured clinical interviews and validated psychometric instruments were administered by trained clinicians. Attendance and homework adherence were monitored. Adverse events were recorded according to institutional ethical procedures.

Relapse verification

Urine toxicology screening was performed at each follow-up point. Positive results were considered indicative of relapse and cross-checked with self-reports.

Statistical analysis

Data were analyzed using IBM SPSS Statistics for Windows, Version 26.0 (IBM Corp., Armonk, NY, USA). Continuous variables are presented as means $\pm$ standard deviations (SD), and categorical variables as frequencies and percentages.

Between-Group and Within-Group comparisons

- Between-group differences in categorical variables were assessed using chi-square (χ^2) tests.
- Independent-samples t-tests were used for continuous variables.
- Within-group changes over time were evaluated using paired-samples t-tests.

Time-to-Relapse analysis

Time-to-relapse across the 6-month follow-up was examined using Kaplan–Meier survival curves and Cox proportional hazards regression. The Cox model included sociodemographic variables, Addiction Severity Index (ASI) domain scores, and clinically relevant covariates such as session attendance, homework adherence, number of prior detoxification attempts, and duration of opioid use. Treatment adherence was dichotomized as good versus poor, defined as attendance of $> 60\%$ of sessions and completion of $\geq 70\%$ of assigned homework tasks [11].

Additional variables

The number of previous detoxification attempts was treated as a count variable, and addiction duration was defined as the number of years between the onset of regular opioid use and study enrollment.

Statistical significance

All tests were two-tailed, and a p-value < 0.05 was considered statistically significant.

Ethical considerations

The study protocol was reviewed and approved by the Ethical Committee of Tanta University Hospitals, Tanta, Egypt (approval code: 36264MD66/4/23). Written informed consent was obtained from all participants prior to enrollment, after providing detailed information on study objectives, procedures, potential risks and benefits, and participant rights, including the right to withdraw at any time without affecting access to ongoing treatment.

Confidentiality was maintained through the use of unique coded identifiers, and all personal data were securely stored and accessible only to the research team. Any adverse events were promptly reported to participants and the ethics committee. Study procedures

adhered to the principles of the Declaration of Helsinki (2013), and all data were used solely for scientific and academic purposes.

Results

Participant flow and retention

A total of 100 male participants with opioid use disorder (OUD) were enrolled. Of these, 90% ($n=90$) completed the 6-month follow-up period. Attrition was 5% ($n=5$) at 3 months and 5% ($n=5$) at 6 months. Figure 1 presents

the CONSORT-style flow diagram for recruitment, allocation, and follow-up.

Baseline sociodemographic and clinical characteristics were comparable between the integrated intervention and control groups (all $p>0.05$; Table 1). There were no significant differences in age, residence, education, marital status, occupation, or history of factors associated with relapse.

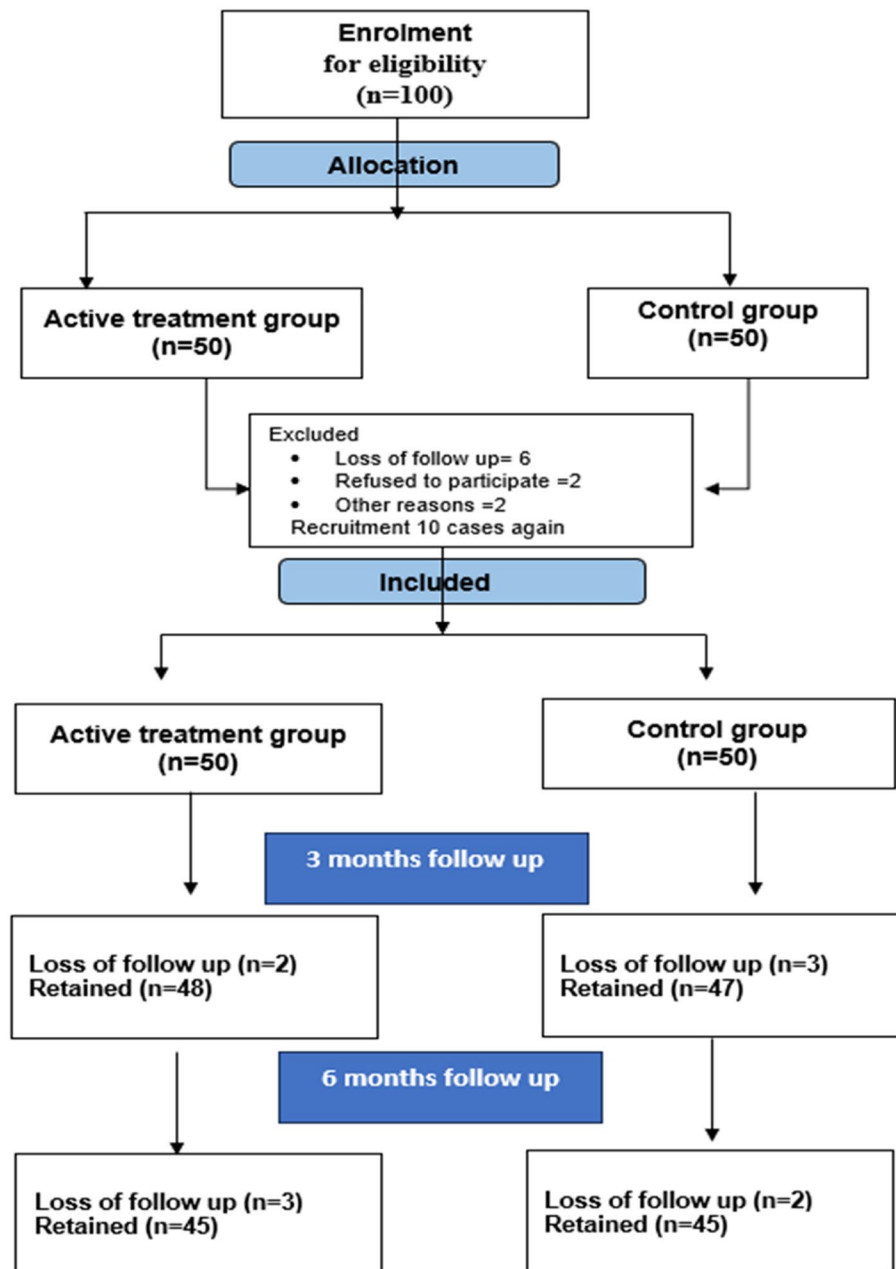


Fig. 1 Flow chart for the study

Table 1 Baseline sociodemographic and clinical characteristics of participants

Variable	Integrated treatment (n=45)	Control (n=45)	Test	p-value
Age (years)	32 ± 9.98	37 ± 8.98	$t=0.89$	0.62
Residence				
Urban	19 (42.2%)	21 (46.7%)	$\chi^2 = 0.87$	0.65
Rural	26 (57.8%)	24 (53.3%)		
Education				
Illiterate	4 (8.9%)	5 (11.1%)	$\chi^2 = 1.43$	0.44
Educated	41 (91.1%)	40 (88.9%)		
Marital status				
Married	22 (48.9%)	16 (35.6%)	$\chi^2 = 1.43$	0.41
Unmarried	23 (51.1%)	29 (64.4%)		
Occupation				
Employed	19 (42.2%)	21 (46.7%)	$\chi^2 = 1.23$	0.41
Unemployed	26 (57.8%)	24 (53.3%)		
History of factors influencing relapse				
Past psychiatric illness	23 (51.1%)	17 (37.8%)	$\chi^2 = 0.67$	0.76
Legal history	17 (37.8%)	15 (33.3%)	$\chi^2 = 0.87$	0.73
Imprisonment	9 (20.0%)	8 (17.8%)	$\chi^2 = 0.21$	0.96
Verbal abuse	17 (37.8%)	15 (33.3%)	$\chi^2 = 0.56$	0.65
Physical abuse	5 (11.1%)	9 (20.0%)	$\chi^2 = 1.14$	0.32
Self-harm/suicide history	11 (24.4%)	17 (37.8%)	$\chi^2 = 0.98$	0.65

(Values are mean ± SD or n (%); no significant differences between groups.)

Table 2 Relapse rate, previous detoxification attempts, and daily opioid dose

Variable	Integrated treatment (n=45)	Control (n=45)	χ^2 (df)	p-value
Relapse after 3 months	15 (33.3%)	30 (66.7%)	4.87 (1)	0.001*
Relapse after 6 months	20 (44.4%)	35 (77.8%)	5.82 (1)	0.001*
Previous detox attempts			1.54 (2)	0.21
0 attempts	8 (17.8%)	7 (15.6%)		
1 attempt	23 (51.1%)	23 (51.1%)		
≥ 2 attempts	14 (31.1%)	15 (33.3%)		
Daily dose (g/day)			0.64 (1)	0.71
< 1 g	22 (48.9%)	24 (53.3%)		
≥ 1 g	23 (51.1%)	21 (46.7%)		

Note. Relapse was confirmed by urine testing. χ^2 = chi-square test for independence; g/day=grams per day. * $p \leq 0.05$ considered statistically significant

Relapse outcomes

At 3 months, relapse confirmed by urine toxicology occurred in 33.3% of the integrated treatment group versus 66.7% of controls ($\chi^2 = 4.87$, $p=0.001$). At 6 months, relapse remained significantly lower in the integrated group (44.4% vs. 77.8%; $\chi^2 = 5.82$, $p=0.001$). There were

no significant between-group differences in prior detoxification attempts or daily opioid dose (Table 2).

Addiction severity index (ASI) outcomes

Baseline ASI scores did not differ between groups. After 6 months, the integrated intervention group showed significant reductions across all ASI domains ($p < 0.001$), whereas the control group demonstrated no meaningful changes (Table 3).

Quality of life (WHOQOL-BREF) outcomes

Baseline QoL scores did not differ significantly between groups. Over 6 months, the integrated treatment group showed substantial and sustained improvements across all QoL domains, while the control group showed no meaningful change (Table 4). Within the integrated group, improvement from baseline to 3 months ($p=0.001$) and from 3 to 6 months ($p < 0.05$) was significant across domains. Among relapses, only the environmental domain improved significantly.

Predictors of relapse

Cox proportional hazards regression identified several independent predictors of 6-month relapse (Table 5). Participation in the integrated treatment group significantly reduced relapse risk (HR=0.65, 95% CI=0.12–0.74, $p=0.012$). Conversely, unemployment (HR=4.98, $p=0.003$), illiteracy (HR=3.43, $p=0.030$), unmarried status (HR=3.86, $p=0.027$), ≥ 2 prior detoxification attempts (HR=3.00, $p=0.015$), daily opioid dose ≥ 1 g (HR=1.49, $p=0.040$), longer addiction duration (HR=1.08 per year, $p=0.050$), poor session attendance (HR=2.43, $p=0.037$), and poor adherence (HR=3.53, $p=0.032$) predicted higher relapse risk. None of the ASI subscales independently predicted relapse, suggesting that psychosocial and behavioral factors outweighed baseline clinical severity.

Discussion

Relapses following detoxification remains a predominant challenge in the treatment of opioid use disorder (OUD). This study demonstrates that a culturally adapted, multimodal psychotherapeutic program integrating cognitive-behavioral therapy (CBT), motivational interviewing (MI), mindfulness-based cognitive therapy (MBCT), family psychoeducation, and peer support yielded a substantial and clinically significant reduction in relapse rates compared to treatment-as-usual (TAU). The near-halving of relapse rates at both three- and six-month follow-ups underscores the capacity of structured psychosocial interventions to foster robust early recovery in inpatient populations. These findings affirm both the feasibility and therapeutic efficacy of implementing

Table 3 ASI domain scores at baseline and 6 months

ASI Domain	Integrated treatment (n = 45)		P	Control group (n = 45)		P
	Baseline	After six months		Baseline	After six months	
ASI – medical	6.09 ± 1.76	1.20 ± 1.30	0.001*	5.44 ± 1.58	2.98 ± 1.56	0.43
ASI – legal	2.22 ± 0.88	0.90 ± 0.75	0.001*	2.11 ± 0.96	2.76 ± 1.52	0.85
ASI – occupational	5.38 ± 2.41	1.80 ± 1.40	0.001*	4.87 ± 2.26	3.29 ± 1.50	0.67
ASI – substance	7.28 ± 1.44	2.50 ± 0.90	0.001*	7.58 ± 1.03	5.04 ± 0.95	0.62
ASI – social	5.60 ± 2.02	1.60 ± 1.10	0.001*	5.49 ± 2.18	6.31 ± 1.38	0.54
ASI – psychiatric	3.78 ± 1.41	1.30 ± 1.20	0.001*	3.60 ± 1.42	3.76 ± 1.52	0.87
Total	4.42 ± 2.22	1.55 ± 1.28	0.001*	4.48 ± 2.21	3.29 ± 1.41	0.76

Values are mean ± SD. Paired-samples t tests; $p \leq 0.05$ significant

Table 4 QOL in integrated treatment and control group, relapse cases and in abstinence cases before and after three months and six months

	Before		Three months after		Six months after		P ₁	P ₂
	Integrated treatment group	Control group	Integrated treatment group	Control group	Integrated treatment group	Control group		
Physical health	44.55 ± 13.9	45.35 ± 17.9	70.40 ± 11.5†	49.22 ± 14.9	80.60 ± 10.5#	50.22 ± 14.9	0.001*	0.03*
Mental health	46.42 ± 15.3	45.05 ± 15.1	71.09 ± 11.8†	50.20 ± 15.9	81.15 ± 9.8#	52.20 ± 15.9	0.001*	0.02*
Social interaction	43.35 ± 14.6	48.35 ± 17.1	68.25 ± 11.6†	48.09 ± 13.1	82.40 ± 10.5#	47.09 ± 13.1	0.001*	0.04*
Environmental quality	44.82 ± 15.2	46.29 ± 15.0	74.07 ± 11.2†	51.24 ± 15.4	84.22 ± 8.7#	53.24 ± 15.4	0.001*	0.03*

QOL: quality of life

Data are presented as mean ± SD. *: Statistically significant at $P \leq 0.05$)

Paired sample Student's T- Test

P₁: Statistically significant at $P \leq 0.05$ between Integrated treatment versus Control group at 3 months

P₂: Statistically significant at $P \leq 0.05$ between Integrated treatment versus Control group at 6 months

†: P – value of comparison between baseline and 3 months in Active treatment

#: P – value of comparison between 3 months and 6 months in Active treatment

Table 5 Multivariate Cox proportional hazards regression of predictors of relapse

Predictor	B	HR	95% CI	p-value
Treatment (active vs. control)	−0.50	0.65	0.12–0.74	0.012*
Age (years)	0.06	1.06	0.94–1.19	0.329
Employment (unemployed vs. employed)	1.61	4.98	1.73–14.33	0.003*
Education (illiterate vs. educated)	1.23	3.43	1.12–10.54	0.030*
Marital status (unmarried vs. married)	1.23	3.86	1.15–10.22	0.027*
Previous detox (≥ 2 vs. 0)	1.10	3.00	1.24–7.27	0.015*
Daily dose (≥ 1 g vs. < 1 g)	0.40	1.49	1.02–2.19	0.040*
Addiction duration (per year)	0.08	1.08	1.00–1.18	0.050*
Attendance (poor vs. good)	1.67	2.43	1.76–5.22	0.037*
Adherence (poor vs. good)	1.63	3.53	1.22–9.54	0.032*

HR hazard ratio, CI confidence interval, B regression coefficient. HR > 1 indicates higher relapse risk; HR < 1 indicates protective effect

comprehensive, non-pharmacological interventions within resource-constrained settings.

Comparative effectiveness and underlying mechanisms

The magnitude of relapse reduction observed parallels and, in several respects, exceeds the outcomes reported

in prior trials examining CBT or mindfulness-based therapies alone. Existing evidence has established the role of CBT in disrupting maladaptive cognitions and the utility of mindfulness practices in modulating affective reactivity and craving [12]. The present results extend this literature by demonstrating that the synergistic integration of cognitive restructuring, motivational enhancement, and mindfulness techniques yields superior outcomes to single-modality or unstructured care.

Marked improvements across all ASI domains including substance use, psychiatric, social, and occupational functioning underscore the intervention's capacity to address addiction as a multidimensional disorder rather than a purely behavioral syndrome. This pattern aligns with findings by Shuman et al. [13], who reported that combined CBT and mindfulness interventions elicit more pronounced gains in emotional self-regulation and craving control than pharmacotherapy or motivational interviewing alone.

Moreover, this study demonstrated substantially lower relapse rates than those reported by Gowid et al. for conventional CBT-only [14] or by Karabulut et al. for standard care [15]. These results resonate with those of Hugno et al. [16], who documented relapse reductions to

38–41% with MBCT plus TAU. Taken together, the evidence suggests that multimodal, mechanism-targeted interventions provide a more coherent and effective therapeutic strategy by concurrently addressing affective dysregulation, cognitive distortions, and habitual reinforcement loops that maintain compulsive opioid use.

Alongside reductions in addiction severity, the intervention yielded consistent improvements across all QoL domains. Since these gains were not observed in the TAU group, the results highlight the critical role of structured psychosocial engagement in promoting long-term well-being. These findings are consistent with the work of Vederhus and Manning [17, 18], highlighting that durable abstinence and active social participation are essential determinants of quality-of-life restoration among individuals with OUD. The breadth of improvements observed spanning emotional, social, and functional areas—indicates that the intervention facilitated not only symptom reduction but also deeper rehabilitative and relational transformation [19].

Predictors of relapse: behavioral and social determinants

Program attendance emerged as one of the strongest independent predictors of sustained abstinence. Participants with high attendance were 2.4 times more likely to remain abstinent ($HR=2.43$), echoing the findings of Komasi, Zullig, and Karabulut [15, 20, 21], which collectively demonstrate that continuity of psychosocial engagement is a decisive determinant of treatment success. These results highlight the role of structured therapeutic exposure in strengthening coping skills, consolidating cognitive change, and reinforcing motivation throughout the recovery trajectory.

Equally notable were the predictive effects of unemployment, illiteracy, and unmarried status. These social determinants align closely with Recovery Capital Theory, which posits that social, economic, and educational resources buffer against relapse by enhancing resilience and reducing exposure to instability [22]. Employment and education contribute to daily structure, self-efficacy, and social embeddedness [23], whereas their absence constrains coping capacity and erodes access to supportive networks. Marital stability further provides emotional buffering and shared responsibility, consistent with Social Control Theory, which conceptualizes intimate relational bonds as powerful, informal regulators of behavior [24]. These findings highlight that psychological treatment cannot be disentangled from the broader social ecology of recovery.

Clinical and behavioral predictors

Clinical severity indicators, including multiple prior detoxifications, higher daily opioid dosage, and longer addiction duration, were independently associated

with heightened relapse risk. This pattern is congruent with Abdelnoor et al. [25], who emphasize that repeated detoxification attempts often reflect entrenched addictive trajectories and declining coping efficacy. High opioid consumption and extended use histories likely reflect deeper neuroadaptive changes that intensify craving and compromise executive control. Additionally, poor adherence was associated with a doubling of relapse risk, reinforcing prior evidence that continuity of engagement is indispensable for sustained abstinence. These observations underscore the need for adherence-enhancing strategies such as digital tracking tools, structured follow-up, peer mentorship, and behavioral reinforcement [26].

Theoretical integration and novel contribution

The findings are theoretically coherent with Bandura's Self-Efficacy Theory, which posits that mastery experiences such as consistent participation and skill acquisition increase perceived control and reduce relapse susceptibility. They also align with the Recovery Capital framework, which emphasizes that psychological, social, and structural resources collectively determine recovery trajectories [27].

Importantly, this study represents one of the first empirical evaluations of a culturally adapted, multimodal psychotherapeutic intervention for OUD within an Egyptian university hospital setting, and one of relatively few globally to integrate multiple evidence-based modalities within a single, culturally responsive framework. The results demonstrate that comprehensive nonpharmacological interventions can yield clinically meaningful and functionally significant recovery outcomes even in low- and middle-income countries. This suggests a pathway for bridging high-income evidence-based models with contextualized, culturally sensitive approaches that are both feasible and effective.

Limitations and future directions

Several limitations should be acknowledged:

1. Non-randomized allocation may introduce selection bias, limiting causal inference.
2. All-male inpatient sample reduces generalizability to women or community-based populations.
3. Follow-up duration of six months does not capture long-term relapse trajectories typical of OUD.
4. Self-report and interviewer-administered assessments may be subject to response bias, particularly in unblinded designs.
5. TAU variability was not strictly standardized, making it difficult to isolate the contribution of individual intervention components.

Future research should address these limitations by employing randomized controlled designs, including gender-diverse and community-based samples, extending follow-up periods, and examining neurocognitive and biological correlates of treatment response. Additionally, integrating digital adherence tracking, vocational rehabilitation modules, and scalable delivery methods could further optimize intervention outcomes and enhance long-term recovery.

Conclusions

The integrated, culturally tailored psychotherapeutic program significantly reduced relapse, improved addiction severity, and enhanced QoL in men with OUD. Benefits persisted across multiple functional domains over six months, highlighting the program's holistic and durable impact. By demonstrating feasibility and effectiveness in a resource-limited setting, this study provides a framework for scaling psychosocial interventions for OUD in similar contexts. Recovery requires restoration of self-regulation, social integration, and personal meaning alongside pharmacologic stabilization.

This work uniquely contributes to the field by integrating evidence-based Western modalities with culturally relevant adaptations, offering a model for advancing OUD rehabilitation policy and practice in the Middle East and other low- and middle-income regions.

Abbreviations

ASI	Addiction Severity Index
CBT	Cognitive-Behavioral Therapy
CDC	Centers for Disease Control and Prevention
CI	Confidence Interval
DSM-5	Diagnostic And Statistical Manual of Mental Disorders, 5th Edition
HIV	Human Immunodeficiency Virus
HR	Hazard ratio
MAT	Medication-Assisted Therapy
MBCT	Mindfulness-Based Cognitive Therapy
MI	Motivational Interviewing
OAT	Opioid Agonist Therapy
OUD	Opioid Use Disorder
QOL	Quality of Life
RCTs	Randomized Controlled Trials
SAMHSA	Substance Abuse and Mental Health Services Administration
SUDS	Substance Use Disorders
WHO	World Health Organization
WHOQOL-BREF	World Health Organization Quality of Life-BREF

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Authors' contributions

- Abdelrahman Ali Elghorab: Contributed to material preparation, data collection, and statistical analysis; drafted the first version of the manuscript and participated in subsequent revisions.- Noha Fawzy Fonon: Contributed to material preparation and data collection; participated in the development of the integrated rehabilitation program and in manuscript revision.- Hossam Eldin Fathallah Elsayy: shared in designing the study and critically reviewed and revised the manuscript for important intellectual content.- Mai Abdelraouf Essa: shared in designing the study, contributed to the development of the integrated rehabilitation program and participated in

manuscript revision.- Mohamed Ahmed Abdelhay: Conceived and designed the study; supervised its implementation; contributed to the development of the integrated rehabilitation program; and critically reviewed and approved the final manuscript.

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Data availability

Data is available on reasonable requests from corresponding author.

Declarations

Ethics approval and consent to participate

The study was done after approval from the Ethical Committee Tanta University Hospitals, Tanta, Egypt (approval code: 36264MD66/4/23). An informed written consent was obtained from the patient or relatives of the patients.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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